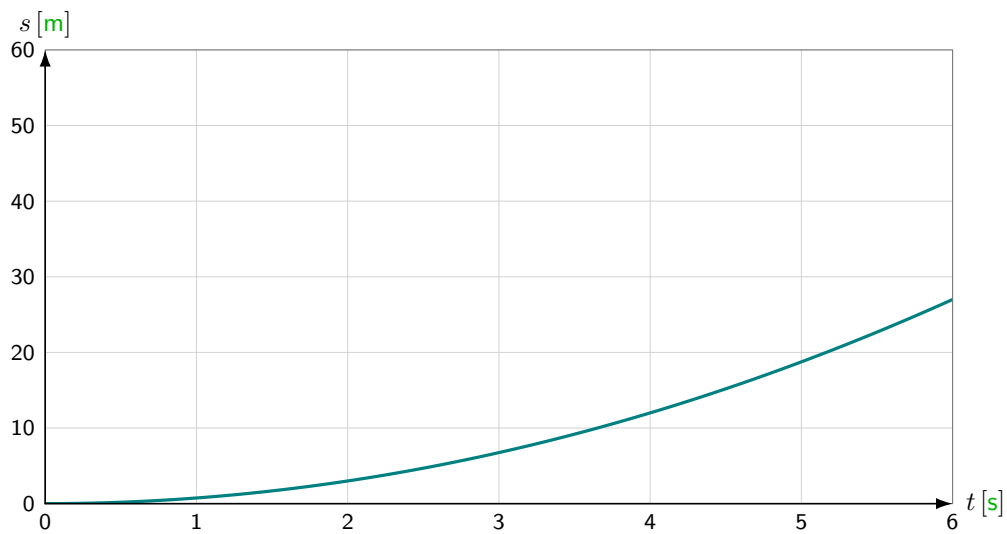


Accelerated motion

A vehicle starts from rest and accelerates uniformly. How are travel time and distance related?

$$s(t) = \frac{1}{2}at^2$$

Change the acceleration with the slider and observe the graph.



1. Read the distance after 2 seconds and after 4 seconds. What factor do you get when you double the time?
2. Double the acceleration from 1.5 to 3 metres per second squared. How does the distance change for the same travel time?

The print version uses the initial acceleration of 1.5 metres per second squared. Use the formula to calculate values for other accelerations.